

Dinh Thai Tuan

📍 Hanoi, Vietnam ✉️ dinhthaituan48@gmail.com in SisypheanHUS 🌐 SisypheanHUS

Profile

Bachelor in Applied Mathematics with experience in alpha research, data analysis, machine learning, and academic research. Interested in quantitative research, machine learning, and factor investing, with a strong background in probability theory, mathematical modelling for data-driven applications.

Education

Vietnam National University – University of Science *Aug. 2022 – Mar. 2027*
Bachelor's degree in Applied Mathematics *(expected)*

- GPA: 2.95/4
- Main courses: Differential Equations, Calculus, PDEs, Probability Theory, Math Modelling, Statistics.

Research and Analytical Experience

Research Consultant *Hanoi, Remote*
WorldQuant BRAIN *Apr. 2024 – Present*

- Developed mathematical models to predict price movements of financial instruments using Fast Expression Language and Python; submitted 1,800+ alphas across global markets (USA, CHN, EUR, GLB, HKG, JPN, AMR, ...).
- Built automated Python pipelines for alpha research: template mining, dataset opportunity mapping, correlation-based diversified selection, and Osmosis/Power Pool allocation via multi-factor scoring under correlation constraints.
- 4-time Grandmaster (top 2%) and 2-time Master (top 8%) in the Genius Program, out of 10,000+ consultants.
- Authored community guides on the BRAIN forum covering Power Pool construction and alpha research methodology.
- Chosen as the exclusive delegate from Vietnam – among 12 representatives selected globally – to attend the International Quant Championship 2025 finals in Singapore, engaging with consultants from around the world.

Research Intern *Hanoi, Vietnam*
Institute of Physics, Vietnam Academy of Science and Technology *July 2025 – Present*

- Proposed a deep learning-based approach for quantifying methanol concentration in ethanol-methanol mixtures using Raman spectroscopy.
- Proposed and evaluated 1D and 2D CNN-based methods for the classification of methanol and ethanol from Raman spectral data.
- First author of a paper published in the proceedings of ICAEP-9 (2025).

Summer Research Student *Hanoi, Vietnam*
Vietnam Institute for Advanced Study in Mathematics *Aug. – Nov. 2025*

- Research in epidemic modeling and actuarial science under the supervision of Dr. Hoang Manh Tuan.
- Contributor of a manuscript on a modified SIS model for health insurance pricing.

Summer Research Student *Nara, Japan*
Group of Mathematical Biology, Nara Women's University *Aug. 2025*

- Short summer research program on dynamical systems in population biology.
- Developed and presented a SIR modeling project in R.

Professional Experience

AI Engineer OJT *Hanoi, Vietnam*
VinSmart Future *Apr. 2026 – Present*

- Studying and evaluating machine learning approaches for fraud detection, including XGBoost and Random Forest, using the PaySim dataset.

Skills

Programming Languages: Python, R, Fast Expression Language, LaTeX

Libraries: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas

Knowledge Areas: Portfolio Construction, Backtesting, Factor Analysis, Machine Learning, Deep Learning, Feature Engineering, Model Validation

Language: English (TOEIC 890 – L/R 460/430)

Awards and Distinctions

- 1st prize, International Quant Championship 2024 – University Round (37,000+ participants)
- 9th place, International Quant Championship 2024 – National/Regional Round

Publications

- **Dinh, T. T.** & Duong, V. (2025). Raman spectroscopy for methanol detection: 1D vs. 2D CNN performance evaluation. *Proceedings of the 9th International Conference on Applied and Engineering Physics (ICAEP-9)*.

Academic Activities

- Summer School on Mathematical Biology 2025, VIASM (30 July – 8 August 2025)
- Samsung Innovation Campus, Artificial Intelligence Course (2024)